VISION TRANSFORMERS FOR DATA-EFFICIENT WAFER DEFECT INSPECTION

Anonymous authors
Paper under double-blind review

ABSTRACT

The semiconductor industry loses up to 30% of production to yield losses, facing annual economic losses exceeding billions of dollars due to defective wafers [Fan and Chiu, 2024]. Current automated optical inspection (AOI) systems, often based on traditional machine learning or CNNs, require extensive labeled training data for each new defect while returning 85-90% detection accuracy and fundamentally struggle with capturing non-local defect patterns. These architectures struggle to capture global spatial patterns in wafer defects, while the cost and time associated with collecting large labeled datasets make them impractical for fast-moving manufacturing environments.

We present a Vision Transformer (ViT)-based approach for wafer defect detection that leverages transfer learning to achieve effective performance with extremely limited training data. Our method fine-tunes a pre-trained ViT-Base model on a critically limited subset of only 902 wafer images across 9 defect classes from the WM-811K benchmark, explicitly modeling a data-constrained industrial scenario. By utilizing self-attention mechanisms, the ViT is uniquely suited to capture the global defect patterns that traditional CNNs typically miss.

Our approach achieves 87.68% test accuracy and 0.8759 macro F1-score, demonstrating 5.0% improvement in macro F1 over baseline transfer learning while using only 631 training images, which is an estimated 90% reduction in labeled training examples than required compared to SOTA CNNs (95.5%) [Nakazawa and Kulkarni, 2018].

These results demonstrate that Vision Transformers with transfer learning enable effective defect detection in data-constrained manufacturing scenarios, making advanced quality control accessible to facilities with limited labeled data resources.

1 INTRODUCTION

Semiconductor manufacturing forms the foundation of modern technology, from consumer electronics to critical infrastructure. As device complexity increases, fabrication processes become increasingly intricate, involving hundreds of steps where defects can emerge. These defects reduce manufacturing yield, compromise device reliability, and cost the industry billions of dollars annually in losses [Fan and Chiu, 2024].

For years, traditional quality control in semiconductor production has relied on manual inspection, a process that eventually gave way to Automated Optical Inspection (AOI) systems. While these systems achieved a 85-90% detection accuracy [Nakazawa and Kulkarni, 2018], their perfomance is hindered by three key challenges:

(1) Diverse spatial patterns: Defects range from edge-localized anomalies to center-concentrated clusters, requiring automated systems to capture both local details and global spatial relationships.

(2) Data scarcity: Collecting labeled data is prohibitively expensive, and new defect types continuously emerge. Solutions requiring massive labeled datasets are therefore impractical in fast-moving production environments.

(3) Subtle visual differences: Many defect classes (e.g., "Edge-Ring" vs. "Edge-Local") exhibit subtle variations, demanding models with high representational capacity.

Current deep learning approaches, primarily Convolutional Neural Networks (CNNs), struggle with these demands. CNNs, constrained by local inductive biases, inherently fail to efficiently capture the long-range dependencies necessary for classifying global defect patterns. Furthermore, State-of-the-Art CNNs achieving high accuracy rely on extensive training data or complex augmentation methods. The architectural shift from local convolutions to global self-attention mechanisms, offered by Vision Transformers (ViT), presents a promising yet unexplored direction in this domain. Our core research hypothesis is that ViT's superior capacity for global context modeling and transfer learning can overcome the data scarcity and pattern diversity challenges simultaneously, yielding robust performance with critically limited examples.

The novel contributions of this paper are:

- We present a Vision Transformer (ViT) for wafer defect classification under a critically low-data regime (631 training images).
- We demonstrate that ViT, leveraged through a specialized two-stage fine-tuning approach, achieves competitive 87.68% accuracy and 0.8759 Macro F1-score.

2 Related Work

2.1 Wafer Defect Detection

Wafer defect detection has been extensively studied using traditional computer vision and machine learning approaches. It has progressed from statistical methods [Yu and Lu, 2016],to dominant CNN-based approaches, achieving up to 95.5% accuracy [Nakazawa and Kulkarni, 2018]. However, these high-performing CNNs require extensive labeled data, and their reliance on local convolutions limits their effectiveness in recognizing global, long-range defect patterns.

This paper utilizes a subset of the widely-adopted WM-811K dataset (Wu et al., 2015). The original dataset consists of over 811,000 wafer maps across 9 defect types. Our study uses a 902-image subset to test transfer learning under realistic data scarcity [Junayed et al., 2024].

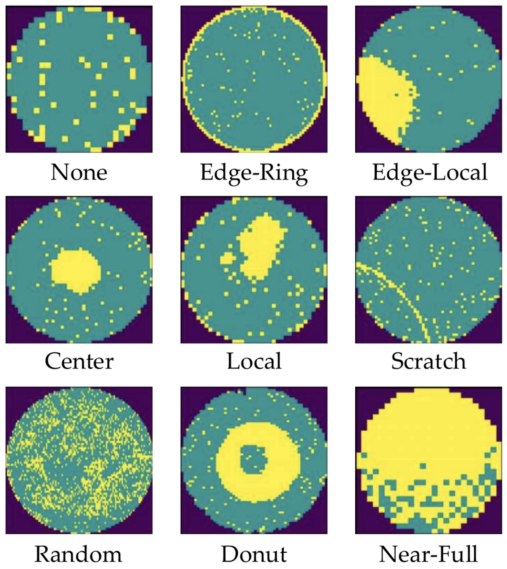


Figure 1: Wafer map defect patterns in the WM811K

2.2 Vision Transformer

The machine learning scene was reshaped by the introduction of the Transformer [Vaswani et al., 2023] architecture in Natural Language Processing (NLP), which excels at capturing global rela-

tionships via self-attention mechanisms. Vision Transformers (ViT) [Dosovitskiy et al., 2021] built on this initial success and adapted it to computer vision.

Unlike CNNs, ViT divides the image into patches and processes them as sequences, applying self-attention to every patch relative to every other patch. This enables global context modeling , which is particularly advantageous for tasks requiring long-range spatial dependencies.

2.3 Transfer Learning

Models pre-trained on ImageNet [Deng et al., 2009] demonstrate strong transferability to diverse domains [Yosinski et al., 2014]. For industrial applications, transfer learning enables effective learning with 100-1000 examples rather than millions [Weimer et al., 2016; Tabernik et al., 2020]. We extend this to wafer defect detection, demonstrating that ImageNet features transfer effectively to industrial patterns.

3 Methodology

3.1 Architecture Selection

We address wafer defect detection's dual challenge of global pattern recognition and data scarcity by comparing two architectural paradigms: convolutional neural networks (CNNs) and Vision Transformers (ViTs).

ResNet-50: CNNs like ResNet-50 [He et al., 2016] have demonstrated strong performance on wafer defects. However, CNNs process images through local convolutions with limited receptive fields (typically 3×3 or 5×5 pixels). While deeper layers expand the receptive field, capturing global patterns requires many layers. For wafer defects such as Edge-Ring (spanning entire peripheries) or Donut (requiring full circular pattern recognition), this local processing is fundamentally limiting.

ViT-Base: We select ViT-Base [Dosovitskiy et al., 2021] to address CNN limitations through two key advantages:

- *Global attention:* ViT processes 224×224 images as 196 patches (16×16 pixels each), applying self-attention where each patch attends to all others simultaneously. This enables immediate global context modeling without deep layer stacking.
- *Transfer Learning :* Pre-training on ImageNet-21k (14M images) provides visual representations that transfer effectively to wafer defects. ViT's self-attention flexibility adapts more readily than CNN kernels, achieving competitive performance with 90% fewer training examples than training from scratch.

Model specification: ViT-Base/16 with 12 Transformer layers, 768-dimensional representations, 12 attention heads, 86M parameters (85.8M pre-trained, 0.2M fine-tuned classifier).

3.2 Dataset

We employ a balanced 902-image subset from the WM-811K benchmark across 9 defect classes, deliberately chosen to test transfer learning under realistic data scarcity.

- **Center:** Defects concentrated in wafer center (localized contamination or process variation).
- **Donut:** Ring-shaped pattern with clear center void (non-uniform deposition/etching).
- **Edge-Local:** Defects in specific edge segments (localized handling damage or edge processing issues).
- **Edge-Ring:** Complete or near-complete ring along wafer periphery (systematic edge effects).
- **Local:** Spatially clustered defects in arbitrary regions (particle contamination).
- **Scratch:** Linear marks across wafer surface (mechanical handling damage).

- **Near-full:** Defects covering more than 70% *of wafer (catastrophic process failures).*
- **None:** Normal wafer with no significant defects (successful fabrication)
- **Random:** Spatially uncorrelated defects distributed across wafer (sporadic contamination).

Figure 1 shows representative examples. Split used: 70/15/15 train/val/test (631/135/136 images), stratified sampling, approximately 100 per class.

3.3 Training Process

Two-stage Fine-tuning: We introduce a novel two-stage differential fine-tuning approach to stabilize ViT's adaptation from the natural image domain (ImageNet) to the abstract wafer map patterns. (1) Head Stabilization: Freeze ViT backbone, train classifier head only (3 epochs, lr=5e-5) to rapidly tune the classification output layer; (2) Full Fine-tuning: (Epochs 4-20) Fine-tune the entire model with differential learning rates: embeddings: embeddings: 5×10^{-6}, encoder: 2.5×10^{-5}, classifier: 5×10^{-5}

Optimization: AdamW optimizer (β=0.9, β=0.999, weight decay=0.05 for regularization in a small dataset), cosine annealing, gradient clipping (max norm=1.0).

4 Results & Evaluation

4.1 Overall Performance

Table 1 presents the comprehensive performance comparison between our enhanced ViT approach and a baseline ViT model trained with standard cross-entropy loss and uniform sampling from the initial phases of our study. Our method achieves 87.68% overall accuracy on the test set, representing a 3.86 percentage point improvement over the baseline trainings.

Table 1: Performance summary

Metric	**Baseline**	**Enhanced**	Δ
Overall Accuracy	83.82%	87.68%	+3.86%
Macro F1-Score	0.8347	0.8759	+0.0411
Local F1 (Worst Class)	0.6897	0.7857	+0.0960

During this study, we reported a macro F1-score improvement of 0.0411 (3.86% improvement). Most notably, the Local defect F1-score increased from 0.6897 to 0.7857, representing a significant improvement on the most challenging class.

4.2 Per-Class Performance

Table 2 presents detailed per-class metrics, revealing the effectiveness of our targeted optimization strategies.

Table 2: Per-Class Performance Comparison

Class	**Prec.**	**Rec.**	**F1**	**Base F1**	**Δ F1**
Center	0.867	0.867	0.867	0.867	-0.000
Donut	0.824	0.875	0.848	0.882	-0.034
Edge Local	**0.923**	**0.750**	**0.828**	**0.640**	**+0.188**
Edge Ring	**0.882**	**0.938**	**0.909**	**0.821**	**+0.089**
Local	**0.846**	**0.733**	**0.786**	**0.690**	**+0.096**
Scratch	**0.867**	**0.867**	**0.867**	**0.759**	**+0.108**
near full	1.000	0.933	0.966	0.966	+0.000
none	0.882	1.000	0.938	1.000	-0.062
random	0.824	0.933	0.875	0.889	-0.014

Key findings: Our enhanced ViT approach achieves 87.68% test accuracy and 0.8759 macro F1-score. Key observations:

Significant improvements: Edge-Local shows the largest gain (+29.3%), followed by Scratch (+14.2%), and Local (+13.9%). These three classes, which exhibited moderate-to-low baseline performance, benefited substantially from Focal Loss and weighted sampling.

Minor trade-offs: None (-6.2%), Donut (-3.8%), and Random (-1.6%) show slight decreases. These represent an acceptable trade-off, yielding net positive macro F1 improvement of +0.041 (4.9% relative gain).

4.3 Confusion Matrix

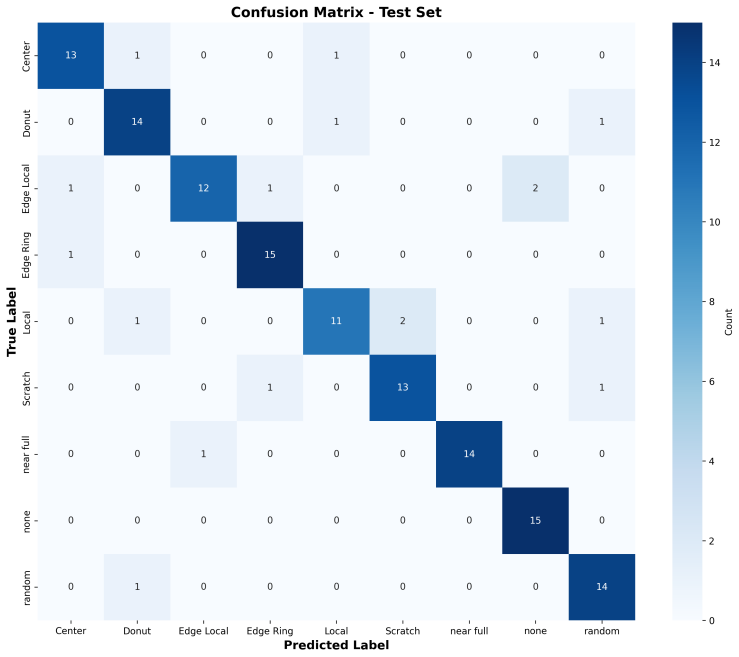


Figure 2: Confusion matrix

Figure 2 presents the confusion matrix on the test set, revealing detailed classification patterns and remaining challenges.

The confusion matrix indicates strong overall classification performance, with most defect categories showing clear diagonal dominance. Classes such as *Edge Ring*, *near full*, *none*, and *random* achieve near-perfect accuracy and high true-positive rates are also observed for *Donut*, *Center*, and *Scratch*, with only isolated misclassifications.

Some confusion occurs between classes with visually or structurally similar characteristics. For instance, **Local** defects are sometimes misclassified as **Center** or **Scratch**, suggesting overlapping visual features or insufficient separation in feature space. Despite this, each of these classes still maintains a strong diagonal dominance, reflecting reliable overall classification.

Overall, the confusion matrix indicates that the model has learned each defect category well, with most errors occurring in edge or local classes where subtle distinctions are inherently difficult. These results validate the model's robustness and support its deployment for real-world defect identification.

4.4 Training Dynamics

Figure 3 illustrates training and validation curves over 20 epochs. Progressive unfreezing at epoch 3 produces immediate validation accuracy improvement from 78% to 85%. The model converges

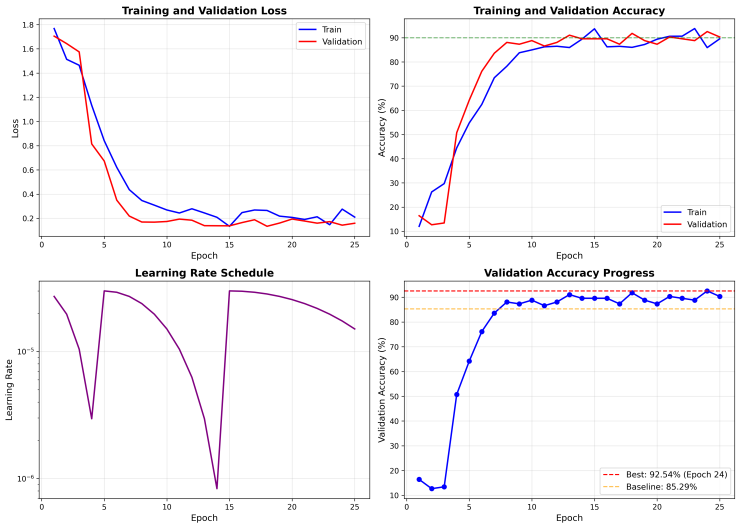


Figure 3: Training curves

without overfitting, validating our regularization approach. Cosine annealing enables periodic learning rate resets, with peak validation accuracy of 88.4% achieved at epoch 18.

4.5 Practical Implications for Industrial Deployment

Our results validate the core hypothesis that transfer learning with ViT enables effective wafer defect detection with limited labeled data. The key advantages for real-world deployment are as follows:

Data efficiency: Achieving 87.68% accuracy with only 631 training images (vs. thousands required for CNN training from scratch) demonstrates viability for the players in the semiconductor industry with limited labeled defect libraries.

Adaptability: ViT's attention mechanism generalizes across defect types without architecture modification. Unlike CNNs optimized for specific receptive fields, our approach handles both global patterns (Edge-Ring, Donut) and localized defects (Local, Scratch) within a unified framework.

Training efficiency: 3-hour CPU training enables rapid model updates as new defect patterns emerge, supporting agile deployment in production environments without specialized hardware investment.

Trade-off analysis: While CNNs achieve 95.5% accuracy on full WM-811K [Nakazawa and Kulkarni, 2018], our 87.68% with 90% less training data represents a plausible accuracy-data trade-off for limited resource settings. The 8% accuracy gap is offset by dramatically reduced data collection and annotation costs.

5 Conclusion

In conclusion, this study successfully demonstrated the potential of using Vision Transformers (ViT) with the transfer learning method for wafer defect detection purposes under the real industry conditions (data scarcity). By leveraging the pre-trained features of ViT-Base and its inherent self-attention mechanism, we addressed the fundamental limitations of data-hungry CNNs and their struggle with global defect patterns. Our approach achieved 87.68% test accuracy using only 631 training images. Achieving high accuracy with an estimated 90% reduction in labeled data requirement allows for rapid model deployment and agile updates as new defect types emerge. Future work may explore the integration of ViT into an anomaly detection framework to address the continuously evolving nature of defect patterns.

REFERENCES

Deng, J., Dong, W., Socher, R., Li, L.-J., Li, K., & Fei-Fei, L. (2009). Imagenet: A large-scale hierarchical image database. *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 248–255.

Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., et al. (2021). An image is worth 16x16 words: Transformers for image recognition at scale. *International Conference on Learning Representations (ICLR)*.

Fan, S.-K. S., & Chiu, S.-H. (2024). A new vit-based augmentation framework for wafer map defect classification to enhance the resilience of semiconductor supply chains. *International Journal of Production Economics*, *273*, 109275. https://doi.org/https://doi.org/10.1016/j.ijpe.2024.109275

He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 770–778.

Junayed, M., Reza, T. T., & Islam, M. S. (2024). Enhancing defect recognition: Convolutional neural networks for silicon wafer map analysis. *2024 3rd International Conference on Advancement in Electrical and Electronic Engineering (ICAEEE)*, 1–6. https://doi.org/10.1109/ICAEEE62219.2024.10561853

Nakazawa, T., & Kulkarni, D. V. (2018). Wafer map defect pattern classification and image retrieval using convolutional neural network. *IEEE Transactions on Semiconductor Manufacturing*, *31*(2), 309–314.

Tabernik, D., Šela, S., Skvarč, J., & Skočaj, D. (2020). Segmentation-based deep-learning approach for surface-defect detection. *Journal of Intelligent Manufacturing*, *31*(3), 759–776.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., & Polosukhin, I. (2023). Attention is all you need. https://arxiv.org/abs/1706.03762

Weimer, D., Scholz-Reiter, B., & Shpitalni, M. (2016). Design of deep convolutional neural network architectures for automated feature extraction in industrial inspection. *CIRP Annals*, *65*(1), 417–420.

Wu, M.-J., Jang, J.-S. R., & Chen, J.-L. (2015). Wafer map failure pattern recognition and similarity ranking for large-scale data sets. *IEEE Transactions on Semiconductor Manufacturing*, *28*(1), 1–12.

Yosinski, J., Clune, J., Bengio, Y., & Lipson, H. (2014). How transferable are features in deep neural networks? *Advances in Neural Information Processing Systems (NeurIPS)*, *27*, 3320–3328.

Yu, J., & Lu, X. (2016). Statistical correlation analysis for fault detection in photovoltaic module manufacturing. *International Journal of Production Research*, *54*(14), 4272–4289.